

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Chemistry

Questions: 181

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

Q. Show that (acv = o. 8 av n, av T, n Sulphur exists in four different (solid rhombic, solid monoclinic, liquid, and gaseous) phases under different conditions. Any two or three phases of sulphur may coexist in equilibrium, but all four phases do not coexist in equilibrium. Give reasons. 8 A certain chemical reaction completes 20% in 15 min at 27°C and in 5 min at 37°C. Calculate the activation energy (in SI units) of the reaction. 8

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2. [UNKNOWN] paper-i

Q1- (a) NaCl (molecular weight : 58.5) consists of a face-centred cubic lattice of Na⁺ ions interlocked with a similar lattice of Cl⁻ ions and has a density of 2.17 g/cm³. Draw the unit cell structure of NaCl. Calculate the number of Na⁺ and Cl⁻ ions that are present in a unit cell. Calculate the volume of the unit cell. The first-order reflection from the d₁₀₀ planes of NaCl occurs at 5.9°. Calculate the wavelength of X-ray. 5+5+5+5=20 For the reaction A → B + C, the following data were obtained: t in sec 900 1800 conc. of A 50.8 19.7 7.62 Prove that the reaction is of the first order. 5 What would be the value of the principal quantum number, if an electron in a hydrogen atom was in the orbital of energy - 0.242 × 10⁻¹⁸ J? Given : k = 2-179 × 10¹⁸ J. 5 Calculate the work done when 1 mole of He expands isothermally and reversibly from a volume of 1 litre to a volume of 10 litres at 25°C. 10

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3. [UNKNOWN] paper-i

Q2. (a) Differentiate between the following: Dropwise condensation and Film condensation Space and Surface resistance Forced and Free convection Black body and Grey body Conduction and Convection 15 A liquid mixture of benzene and toluene containing 50 mole percent benzene is to be separated in a distillation column to achieve 98 mole percent benzene in distillate and 2 mole percent benzene in residue. The feed is a saturated liquid. If the reflux ratio of 4.0 is to be used, find the number of theoretical plates needed for a feed rate of 100 kmol/hr by McCabe-Thiele method. Use relative volatility of benzene-toluene = 2.16. Also, calculate the production rates. 15 Explain the batch sedimentation process showing different zones formed at various stages of settling. 10

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4. [UNKNOWN] paper-i

Q2. (a) Using de Broglie wave-particle dual relationship, $\lambda \times p = h$; calculate the wavelength (in SI units) of (i) an electron travelling 1.00% of the speed of light, and (ii) a baseball (mass = 140 g) travelling at 40 ms⁻¹. Comment on your findings. 15 (b) (i) A non-ideal gas is used in a Carnot engine. Find the values of AU, AH, AS_{univ}, AA, AG for a cycle. Give reason's. At constant pressure, for fixed amount of a substance, with increase of temperature enthalpy (H) of the substance increases but Gibbs free energy (G) decreases. Give reasons. At constant pressure and temperature, adsorption of gas on solid surface occurs spontaneously. What will be the signs of AH, AS and AG of this process? 5+5+5=15 (c) Calculate the number of photons in 10 kcal of light of wavelength 350 nm. 10 2 ===== PAGE 3 =====

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5. [UNKNOWN] paper-i

Q2. (a) If uncertainty in position is written as Δx and in momentum as Δp , then Heisenberg Uncertainty principle is $\Delta x \Delta p \geq \frac{h}{4\pi}$. If the position of an electron is known to within 10⁻¹² m, what is the uncertainty in its momentum? Given $h = 6.626 \times 10^{-34}$ Js and $m_e = 9.1 \times 10^{-31}$ kg. 5 (b) One mole of water is vapourised reversibly at 100°C and 1 atm. H₂O (l) → H₂O (g) The heat of vapourisation of water is 9720 cal/mol. Calculate W, AE, AH and AS. 20 ===== PAGE 3 ===== How is molecular partition function defined? What is the physical significance of this property? Discuss the effect of temperature on the molecular partition function. 10 Write brief notes on n-type and p-type semiconductors. 5

6. [UNKNOWN] paper-i

Q3. (a) Explain the working of rotary vacuum drum filter with the help of neat sketch. 10 (b) Suppose a 3 inch schedule 80 pipe (inside diameter = 73.66 mm; outside diameter = 88.9 mm) having thermal conductivity of 43 is covered with 25.4 mm of an insulation having thermal conductivity of 0.06 m°C. The outside of the insulation is exposed to an environment having heat transfer coefficient of 10 and temperature is 20°C. The inside temperature of the pipe is 250°C. Calculate the overall thermal resistance and heat loss for unit length of the pipe with insulation. 15 3 ===== PAGE 4 ===== (c) Oxalic acid is to be crystallized from a saturated aqueous solution initially at 80°C. To what temperature does the solution have to be cooled to crystallize 95% of the oxalic acid as the dihydrate? 151 Solubility data vs Temperature of oxalic acid Temp °C 0 10 20 40 60 80 Solubility c gms/100 gmol H2O 3-5 6 9-5 21.6 44.3 84.4

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7. [UNKNOWN] paper-i

Q3. (a) At 27°C, for fixed amount of (i) ideal gas, (ii) hydrogen gas, (iii) ammonia gas; show the plots of 'Z' (compressibility factor) against P (pressure) on the same diagram. 10 Derive the expression of half-life period ($t_{1/2}$) of a first order chemical reaction from its integrated rate equation. Does this value depend on the temperature of the reaction? Give reason in support of your answer. 15 (i) Using Stirling's approximation, we write $\ln(n!)$ $\approx n \ln n - n$. Calculate the % of error when (A) $n = 10$ and (B) $n = 69$. Comment on your results. (ii) Calculate the interplanar distance between two successive 123-Miller planes of an orthorhombic crystal with unit cell lengths $a = 0.82$ nm, $b = 0.94$ nm, and $c = 0.75$ nm. 7+8=15

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8. [UNKNOWN] paper-i

Q3. (a) Calculate the mean activity coefficient at 25°C of (i) 0.01 molal solution of LiCl , and (ii) 0.001 molal solution of BaCl_2 . Given : $\gamma_{\pm} = 0.509$ for water at 25°C. 10 (b) Consider the Arrhenius equation. Derive the expression relating rate-constant, energy of activation and frequency factor, in the form of a straight line equation. 10 (c) According to Van der Waals' equation, calculate the pressure required to confine one mole of CO_2 in a volume of 1 litre at 0°C. 10 Given: $R = 0.082$ litre atm / mol K, $a = 3.60$ atm litre²/mol², $b = 4.27 \times 10^{-2}$ litre/mol (d) For the photochemical reaction it is found that 1.00×10^{-5} mole of B is formed, as a result of the absorption of 6.00×10^7 ergs at 3600 Å. Calculate the quantum yield. 10 Given: Avogadro number 6.02×10^{23} Planck's constant 6.626×10^{-34} J-sec Velocity of light 3×10^{10} cm/sec

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9. [UNKNOWN] paper-i

Q4. (a) Explain briefly the following terms: 10 Relative humidity Adiabatic saturation temperature Wet bulb temperature Equilibrium moisture content HETP (b) A shell and tube heat exchanger with one shell pass and two tube passes is used to heat 5 kg/s of water from 30°C to 80°C. The water flows in the tubes and condensing steam at 1 atm is used in the shell side. Calculate the area of the heat exchanger, if overall heat transfer coefficient is

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10. [UNKNOWN] paper-i

Q4. (a) The equilibrium constant, K_p of a certain gas-phase reaction, varies with temperature (T in K) as $\ln K_p = -\frac{104}{T} + \text{constant}$. The temperature range of 300 to 500 K. 1088K $1.51 \times 10^5 \text{ K}^2$. the $+ T^2$, in of the reaction ($\Delta_r S^\circ$) at 400 K. Calculate the standard entropy change Given : $\Delta_r G^\circ = -RT \ln K_p$ and $\Delta_r S^\circ = R \ln K_p$ 15 L aT jP Phosphorescence emission rate is slower than fluorescence emission 10 rate. Explain with Jablonski diagram. What are the respective electrode reactions and cell reaction of the following galvanic cell? $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) || \text{Ag}^+(\text{aq}) | \text{Ag(s)}$. Calculate the equilibrium constant value of the cell reaction at 25°C. Given: $E^\circ_{\text{H}^+/\text{H}_2} = 0.7996$ V at 25°C. 15 F.51-P-C/tM 3 ===== PAGE 4 ===== SECTION B

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11. [UNKNOWN] paper-i

Q4. (a) Considering molecular-orbital energy level diagram, justify the σ - σ^* bond distances in O_2 , O_2^+ and O_2^- as 121, 128

and 149 A, respectively. 10 Draw and discuss the pressure-temperature diagram for H₂O. Apply the phase rule to the diagram 10 Calculate the equilibrium constant for the following reaction at 25°C : $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$ 10 Given at 25°C: $e(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$ and $e(\text{Cu}^{2+}/\text{Cu}) = 0.34 \text{ V}$ FSI-P-C1-1-A4 3 ===== PAGE 4 ===== (d) Consider the reaction : $\text{PC}_{15}(\text{g}) \rightleftharpoons \text{PC}_{13}(\text{g}) + \text{C}_{12}(\text{g})$ Derive the expression relating K_p and degree of dissociation α . Given : At 250°C and 1 atm, K_p for the above reaction is 78. Calculate α . 8+2=10 FSI-P-C1--fM 4 ===== PAGE 5 ===== SECTION B

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12. [UNKNOWN] paper-i

Q5. (a) List the important characteristics to be considered while selecting a material of construction. 5 (b) Define the following: Flux Permeability Trans-membrane pressure Molecular weight cut-off 5 (c) Estimate the thickness of the cylindrical vessel having 1.5 m diameter. The vessel is to operate at a pressure of 15 bar absolute and temperature of 300°C. 5 Given — MOC : Steel (Design stress = 85 N/mm²); A corrosion allowance of 3 mm and Joint efficiency (J) = FO. (d) What are Horton spheres? Give its applications. 5 (e) Define controller's gain, I_c and discuss the effect of value of I_c on the offset. 5 (f) Find the transfer function $Y(s)/X(s)$ of the system shown in the figure. 5 1 0.5 T S 1 (g) List the 'devices used for high temperature measurement in industry giving their applications. 5 x y FSI-P-CHME 6 ===== PAGE 7 ===== (h) A step change of magnitude 40 is introduced into a system having (. Y s) 10 transfer function Determine : Overshoot Maximum value of $y(t)$ Ultimate value of $y(t)$

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13. [UNKNOWN] paper-i

Q5. (a) How many isomers are possible for $[\text{Co}(\text{NH}_3)_6(\text{NO}_2)]$ C12 ? Draw their structures and mention their names. 8 (b) What are labile and non-labile complexes ? Identify labile and non-labile complexes from the following: 8 $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ $[\text{Co}(\text{NH}_3)_5(1120)]^{3+}$ (c) Identify the intermediate and the product in the following reaction sequence : 8 $\text{PhLi} [\text{Me}_3\text{O}^+ \text{BF}_4^-] \text{W}(\text{CO})_6 > \text{Intermediate} \rightarrow \text{Product LiBF}_4, \text{Me}_2\text{O}$ (d) Although N₂ is available in atmosphere in plenty (- 78% by volume), it does not react with some of the reactive metals easily. Give three reasons for inert nature of N₂ towards the metals mentioned above. 8 (e) What is a super acid ? How can one prepare super acid ? The super acid upon reaction with neopentane gives an intermediate that on further decomposition afforded Me₃C⁺ and CH₄. Identify the intermediate. 8

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14. [UNKNOWN] paper-i

Q5. (a) Draw the structures and d-orbital splitting diagrams of (i) $[\text{NiCl}_4]^{2-}$, and (ii) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$. Calculate their crystal-field stabilization energy (CFSE) and spin-only magnetic moment values. 10+10=20 (b) Draw the structures of the proteins (i) de-oxy myoglobin, and (ii) oxidised form of cytochrome-c. Comment on the properties of de-oxy myoglobin. 5+5+10=20

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15. [UNKNOWN] paper-i

Q6. (a) Discuss the algorithm for the design of a skirt support for a tall vertical vessel. 15 Find the value of I_{ce} for a stable system having the characteristic equation $s^3 + 6s^2 + 12s + 6(1 + I_{ce}) = 0$ using Routh Array test. 10 Discuss the different membrane modules used for reverse osmosis. 15 $X(s) = \frac{52 + 1.6s}{s^2 + 4}$ FSI-P-CHME 7 ===== PAGE 8 ===== 1

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16. [UNKNOWN] paper-i

Q6. (a) The magnetic moment of a certain Co(II) octahedral complex is 4.0 μ_B . What is its d-electron configuration? 15 Explain the variation in ν_{CO} (cm⁻¹) of the following isoelectronic metal carbonyls: 15 Compound ν_{CO} (cm⁻¹) NEC 0)4 2060 $[\text{Co}(\text{CO})_4]^-$ 1890 $[\text{Fe}(\text{CO})_4]^{2-}$ 1790 How many geometrical isomers are possible for $[\text{Ru}(\text{H}_2\text{O})_3\text{C}_3\text{H}_3]$? Draw their structures and indicate their names. 10

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17. [UNKNOWN] paper-i

Q6. (a) Explain the structure and bonding in $[\text{Cr}(\text{CO})_6]$ and $[\text{PtCl}_3(\text{C}_2\text{H}_4)]^-$, showing metal-ligand orbital interactions,

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Q7. (a) Construct the Bode plot for the system having $G(s) = \frac{20}{(s + 1)(s + 5)}$ Semi log graph paper UT FSI-P-CHME 8

===== PAGE 9 ===== Log log graph paper MI= MI_MN 1111/ln110111111111111111
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111111111111111111111111 1111111 1111111 1111111 What are the advantages of supercritical fluid separation technique ? List four applications for the same. 10 List the different types of closures used for vessels under internal pressure. Draw neat sketches of the same. On what factors does the choice of closure depend? 10 FSI-P-CHME 9

===== PAGE 10 =====

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Q7. (a) Draw the structures of the following metal carbonyls and show electron counting for each one of the complexes :
 $15 \text{ Co}_2(\text{CO})_8$ $10 \text{ Mn}_2(\text{CO})_{10}$ FS-P-C/t/4 ===== PAGE 5 ===== (b) Draw two resonance structures (A and B) for metal
olefin complex, $\text{LaM}(\text{C}_2\text{X}_4)$. In which of the following complexes does the resonance structure A or B dominate and
why? 15 Rh H H (c) Consider the following equations and explain the observed trend in log K values, en refers to
ethylenediamine, $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$. $10 [\text{Cd}(\text{H}_2\text{O})_6]^{2+} + \text{en} [\text{Cd}(\text{en})]^{2+} + 2 \text{H}_2\text{O}$, $\log K = 5.84$ $[\text{Cd}(\text{H}_2\text{O})_6]^{2+} +$
 $2 \text{N}_3^- \rightleftharpoons [\text{Cd}(\text{NH}_3)_2(\text{H}_2\text{O})_4]^{2+} + 2 \text{H}_2\text{O}$, $\log K = 4.95$.

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Q7. (a) Draw the solid-state structure of $\text{Co}_2(\text{CO})_8$ and show valence-electron count around Co atom. 10 Explain the term 'over potential'. Discuss the application of over potential in (i) electro-deposition of metals from solutions, and (ii) corrosion of metals. 15 Consider the complex $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$. Draw the structures of possible geometrical isomers. 5 (d) Draw the structure of ferrocene. Showing the number of electrons contributed, count the number of valence-electrons around the Fe atom. 10 ===== PAGE 6 =====

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Q8. (a) Describe the working of Bourdon tube pressure gauge with the help of a neat diagram. 10 Explain the principles of ion-exchange method and its application to water softening process. 15 Answer the following in brief: 15 What are the various losses encountered while storing volatile liquids? What are wind girders ? Why are they provided? What is reinforcement ? Why is it provided? FSI-P-CHME 10

22. [UNKNOWN] paper-i

Q8. (a) What are the key points invoked for oxidative addition (OA) and reductive elimination (RE) ? In the given two equations below, which one corresponds to OA and which one corresponds to RE and why? 11 Et.P PEt, Et3P z PEt3 Rh Rh MeC CH Et3P // I \ C PEt3 C C C C \ / \ Me Me Me (1) Cl Ph3P\ Cl Pd Ph / \ Cl PIN — PliCl Cl \ / PPh3 Pd \ Ph3P •Cl 15 ===== PAGE 6 ===== The magnetic moment of La3+ (f°), Gd3+ (f7) and Lu3+ (f14) can be calculated using Isspin only formula. However, for the other lanthanides gspin only formula fails. Explain why. Write down the formula to calculate magnetic moment for those lanthanides other than La3+, Gd3+ and Lu3±. 15 Explain the spin state and electronic configuration of iron(H) in myoglobin and O2 bound myoglobin. How does the geometry of iron(II) in these metalloproteins vary subtly? 10 It FS-P-01-W 6

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23. [UNKNOWN] paper-i

Q8. (a) Comment on the consequences of 'Lanthanide Contraction'. 10 Derive the B.E.T. equation for adsorption on a solid surface. How can the surface area be determined with the help of B.E.T. equation? 10 Discuss the merits and demerits of liquid hydrogen fluoride as a non-aqueous solvent. Give the chemical reactions which take place in this solvent. 10 What do you understand by quantum yield ? Discuss high and low values of quantum yield by taking suitable examples. /0 PSI-P-0/1M 6

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24. [UNKNOWN] paper-i

900 . Suppose, this same heat exchanger is used with entering m2 cic water at 30°C, but with a water flow rate of 1-3 kg/s. What would be the exit water temperature under these conditions, if overall heat transfer coefficient is 700 - ? The specific heat of water may be considered m2°C as 4179 . The relation of effectiveness (s) and number of transfer kg°C units (NTU) may be expressed as $e = 1 - \exp(-NTU)$. 15 FSI-P-CHME 4 ===== PAGE 5 ===== 15 m m Pump (c) Water at 20°C is being pumped from a tank to an elevated tank at the rate of 5 litre/s through a 4 inch schedule 40 pipe (Inside Diameter = 04023 m) as shown in the figure below. Calculate the power required for the pump considering pump efficiency as 65%. 15 Given Data: density of water = 998.2 kg/m3 viscosity of water = 1.005 x 10-3 Pa-s loss coefficient for elbow = 0.75 16 friction factor (f) = (Laminar flow) Reynolds number (NRe) $f = 0.079 \text{ Nit}^{25}$ (4000 < NRe < 105) Neglect contraction and expansion losses. FSI-P-CHME 5 ===== PAGE 6 ===== SECTION B

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25. [UNKNOWN] paper-ii

1. The number of radial nodes for 2s, 3p and 3d orbitals, respectively are: (a) 2, 1, 0 (b) 2, 1, 1 (c) 1, 0, 0 (d) 1, 1, 0

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26. [UNKNOWN] paper-ii

1. The electronegativity values of both the elements are almost same

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27. [UNKNOWN] paper-ii

1. Nitric oxide gas is paramagnetic but liquid nitric oxide is diamagnetic

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28. [UNKNOWN] paper-ii

1. The most probable speed increases with increasing temperature and with decreasing molar mass and simultaneously distribution becomes broader

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29. [UNKNOWN] paper-ii

1. Joule Thomson effect occurs at constant enthalpy

30. [UNKNOWN] paper-ii

1. Higher intake of salt causes higher osmotic pressure

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31. [UNKNOWN] paper-ii

1. Answer the following questions : 5x8=40 Explain why the dipole moment of naphthalene is 0.00 D and that of azulene is 1.02. D. Which one of the following compounds is more acidic? Justify your answer: $\bullet$ and Ph_3CH Write the structure of the product(s) with proper stereochemical outcome of the following reaction and also the corresponding mechanistic pathway : $\text{Ph H Br}_2 \text{ CO}_2\text{H}$ Predict the products of the following reaction showing distribution of the labelled carbon) 14t...a $\bullet$ and the mechanism : O Cl NaOMe Account for the difference in reactivity of the following two reactions based on diastereomeric chlorohydrins. Write the product in each case : $\text{KZL}^{\text{'''}}/\text{C1 O:OH Cl Cold -OH/ H}_2\text{O Slow Cold OH/H}_2\text{O Fast}$ The initial product formed from the reaction of 2-methylcyclohexadiene and dimethylacetylene dicarboxylate on heating is transformed to another product with elimination of a small molecule. Write the structure of each product (initial and final). Also write how these are formed. $\text{FSI—D—CHM/ 44 2 ===== PAGE 3 =====}$ (g) Acetylation of methanol using acetyl chloride and pyridine is proposed to proceed through the following two pathways : (i) Via formation of pyridinium acetate (ii) Via formation of ketene Can you devise an experiment to prove through which pathway the reaction actually proceeds? (h) When irradiated, tetrafluoroethene undergoes a cycloaddition reaction with 1-buten-3-yne to yield two products in almost equal amounts. What are the structures of these products?

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32. [UNKNOWN] paper-ii

1. Answer the following: 5x8=40 1.(a) Name a sandwich compound which is aromatic in character. Draw its structure. 5 What are sydnones ? Draw a representative structure. 5 With a suitable example show 'Norrish Type-II' cleavage reaction. 5 Show the mechanism for the following reaction: $\text{O O CH}_3 \cdot \text{HI P1 I 2113C C — CH}_3 \longrightarrow 113\text{C — C — CH= C — CH}_3 + \text{H}_2\text{O}$ Mention the product in the given reaction: 0 (i) $\text{Zn CH}_3\text{CH}_2\text{C}_{112} \text{ — C —}_{11} + \text{BrCH}_2\text{CO}_2\text{CH}_2\text{CH}_3$ 4> H30 Also provide mechanism of the above reaction. How 2,3-dimethyl-2,3-butanediol can be converted to 3,3-dimethyl-2-butanone ? Suggest the mechanism of the involved reaction. 5 / + OsO₄ $\longrightarrow$ A / $\bullet$ Aqueous $>13 + (\text{H}_2\text{O})_2\text{OsO}_2$ ethanolic NaHSO₃ solution Identify A and B. 5 Explain why di-*t*-butoxycarbene is unreactive. 5 The rate of hydrolysis of $(\text{CH}_3)_3\text{CBr}$ in 50% aqueous methanol is thousand times faster than that in pure methanol — explain. 5 (ii) Mention the product(s) in the following reaction. Elucidate your answer. $\text{CH}_3 \text{ — CH}_2 \text{ I — H — C — heat SME } \longrightarrow ?$ 5 2.(b) Write down the names and the structures of the predominant products for the following elimination reactions : $\text{EtO CHCH}_3 \rightarrow 3 \text{ Br CH}_3\text{CH}_2\text{CHCH}_3$? EtO₈ 10 Me₃ 5 5 fsi-d-cta 2 ===== PAGE 3 ===== 2.(d) Give the mechanistic steps for the synthesis of aniline from benzamide. 5 Why does LiAlH_4 usually fail to reduce a C — C multiple bond ? Why does such a bond in conjugation with an aromatic system on one side and a carbonyl group on the other side get reduced by LiAlH_4 ? 5 Give the mechanistic steps and mark the major product of the following reaction: $\text{CH}_3 \text{ CH}_3 ? 1 \text{ O C}_{113} \text{ — CH — C — 0-13 acid Foic} > \text{Br as solvent}$ 3.(a) Write down the stability order of the following alkyl radicals. Explain your answer. $\text{C}^\bullet\text{H}_3$, $\text{MeC}^\bullet\text{H}_2$, $\text{Me}_2\text{CH}^\bullet$ and $\text{Me}_3\text{C}^\bullet$ 10 3.(b) Which of the following is(are) aromatic? Cyclopropenyl cation Cyclopropenyl anion Cyclopropenyl free radical Justify your answer. 10 3.(c) Account for the observed dipole moment and aromaticity in azulene. 10 3.(d) Cis-2-butene + Singlet carbene ? Cis-2-butene + Triplet carbene $\rightarrow$? Mention the product(s) in each of the above cases. Substantiate your answer. 10 Mention the product(s) in the following reaction. Suggest the mechanism of this reaction. What is the popular name of this reaction? $\text{CH R} \cdot 2 \text{ ZnC}_{12} \text{ NH N= C ? [le, R = alkyl, aryl or hydrogen]}$ 10 Write down the mechanism of the following reaction: hi) Norrish Type-1 10 3 fsi-d-chm ===== PAGE 4 ===== (i) Between [14] Annulene and [16] Annulene — which one is aromatic and why? 5 (n) Write the structure of the product formed in the following reactions : $\text{I. H-1}^+\text{ CH}_2 \text{ I } \rightarrow \text{Cl-I}^+ \text{ Li/Hg}$ 5 Br With a suitable free-energy profile diagram illustrate and explain what is meant by 'thermodynamically controlled product' and 'kinetically controlled product'. 10

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33. [UNKNOWN] paper-ii

2. What fraction of the total number of electrons is in the p-sublevels in Iron (Fe)? (a) 50% (b) 49% (c) 46.15% (d) 47.82%

34. [UNKNOWN] paper-ii

2. Both the elements have almost identical values of their ionic potential Which of the statements given above is/are correct? (a) 1 and 2 (b) 1 only (c) 2 only (d) Neither 1 nor 2

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35. [UNKNOWN] paper-ii

2. Nitric oxide gas is diamagnetic but liquid nitric oxide is paramagnetic

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36. [UNKNOWN] paper-ii

2. The most probable speed increases with increasing temperature and with increasing molar mass and distribution becomes sharp

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37. [UNKNOWN] paper-ii

2. For perfect gas Joule Thomson coefficient is zero

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38. [UNKNOWN] paper-ii

2. It results in greater inflow of water in the cells and causes water retention in the tissue cells and intercellular spaces Which of the statement(s) given above is/are correct? (a) 1 only (b) 2 only (c) 1 and 2 (d) Neither 1 nor 2

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39. [UNKNOWN] paper-ii

Q2. (a) A batch adiabatic reactor at an initial temperature of 373 K is being used for the reaction : $A \rightarrow B$. Assume that the heat of reaction is -1 kJ/mol at 373 K and the heat capacity of both A and B to be constant and equal to 50 J/mol K . Calculate the temperature rise for 1 mole of feed and after a conversion of 0.5. 15 A wet paper pulp contains 75% water. After 100 kg of water is removed in a dryer, it is found that the pulp now is containing 30% water. Calculate the weight of the original pulp. 15 One mole of methane undergoes complete combustion in a stoichiometric amount of air. Both the reactants and products are in the gas phase. Estimate the mole fraction of water vapour in the product gas. 10 FSI-D-CHME 2 ===== PAGE 3 =====

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40. [UNKNOWN] paper-ii

2. (a) Predict the product(s) of the following reactions with proper stereochemical outcome and denote the corresponding pathways (abbreviated) : $\text{SOC}_{12} > 4$; p 113 $\text{SOC}_{12} \text{ Ph OH}$ Explain why methyl alkyl ketones ($\text{RCH}_2\text{COCH}_3$) undergo aldol condensation via attack at the methyl group in the presence of base catalyst, but via attack at the methylene group in the presence of acid catalyst. Write the structures of [A] and [B] of the following sequence of reactions. Write the mechanism of their formation. What type of rearrangement and electronic shift are involved in the second step? $\text{H}_3\text{C} > \text{OCH}_3 \text{ OCH}_3 \text{ C}_{2113}\text{CO}_2\text{H}$ [A] $\rightarrow$ [B] 10 What type of mechanistic pathway is involved in each case of hydrolysis of S, S- and S, R-2-acetoxycyclohexyl tosylate, separately? Which one reacts at a faster rate and why? Write the structure of the product(s) with proper stereochemical outcome and mention R/ S to the chiral carbons. Write the structure of the intermediate also (where applicable). 10 Account for the formation of product(s) from a reaction of 3-bromoanisole with 2 /liq. NH_3 . 5 > ? 5 FSI-D-CHM/ 44 3 [P.T.O. ===== PAGE 4 ===== $\text{BnNH NO}_2 \text{ NO}_2 + \text{BnNH}_2 \rightarrow \text{NO}_2 \text{ BnNH NO}_2 \text{ NO}_2 + \text{BnNH}_2 \text{ NO}_2 \text{ Cl NO}_2 \text{ NO}_2$ (I) Using the same coordinate system, draw two separate energy diagrams for the following two reactions. How can you denote (abbreviated form) such reactions? 5

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41. [UNKNOWN] paper-ii

3. Which is the correct order of degree of hydration among the following ions? (a) $\text{Be}^{2+} > \text{Mg}^{2+} > \text{Sr}^{2+} > \text{Ba}^{2+}$ (b) $\text{Be}^{2+} > \text{Sr}^{2+} > \text{Mg}^{2+} > \text{Ba}^{2+}$ (c) $\text{Ba}^{2+} > \text{Sr}^{2+} > \text{Mg}^{2+} > \text{Be}^{2+}$ (d) $\text{Ba}^{2+} > \text{Mg}^{2+} > \text{Sr}^{2+} > \text{Be}^{2+}$

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42. [UNKNOWN] paper-ii

3. Nitric oxide is a neutral oxide and is not an acid anhydride Which of the statement(s) given above is/are correct? (a) 1 and 3 (b) 1 only (c) 2 and 3 (d) 2 only

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43. [UNKNOWN] paper-ii

3. The end point depends on pH of solution which is controlled by adding ammonia –ammonium citrate buffer

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44. [UNKNOWN] paper-ii

3. The most probable speed decreases with the decrease of temperature and with increasing molar mass and simultaneously distribution becomes sharp Which of the statement(s) given above is/are correct? (a) 1 and 3 (b) 2 and 3 (c) 1 only (d) 2 only

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45. [UNKNOWN] paper-ii

3. Heating effect corresponds to positive value of Joule Thomson coefficient Which of the statements given above is/are correct? (a) 1 and 3 (b) 1 and 2 (c) 2 and 3 (d) 3 only

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46. [UNKNOWN] paper-ii

3. $-\Delta S = (\quad)$ Select the correct answer using the code given below: (a) 1 and 2 only (b) 1 and 3 only (c) 2 and 3 only (d) 1, 2 and 3

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47. [UNKNOWN] paper-ii

Q3. (a) The homogeneous gas phase decomposition of phosphine $4\text{P}_{113}(\text{g}) \rightarrow \text{P}_4(\text{g}) + 6\text{I}_{12}$ proceeds at 922 K with first order rate - $r_{\text{PH}_3} = (10/11r) \text{C}_{\text{PH}_3}$ What size of plug flow reactor operating at 922 K and 460 kPa can produce 80% conversion of a feed consisting of 40 moles of pure phosphine per hour? 15 The rate constant for a second-order reaction at 300 K is 2.5 and the activation energy is 20000 J/mol. What are the units of the rate constant ? At what temperature would the rate be triple of the rate at 300 K for this reaction? 10 A first-order, irreversible liquid-phase reaction is taking place in a CSTR and 50% conversion is obtained. If two more CSTRs of the same size are placed downstream, what is the final conversion? 15

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48. [UNKNOWN] paper-ii

3. (a) Discuss why p-dimethylaminobenzaldehyde does not undergo any reaction with KCN, but a mixture of this and benzaldehyde does, giving a condensation product. Name the reaction also. Predict the product and draw the probable mechanistic pathway of the following reaction : 5 & i) $\text{Mg/Et}_2\text{O}$, A Cl ii) $\text{NH}_4\text{Cl}(\text{aq})$ (c) Write the structures (A to F) for the following sequence of reactions : 10 $\text{CrO}_3 \cdot \text{CH}_3\text{OH} \cdot \text{CN} \cdot \text{CH}_3 \cdot \text{K}_2\text{CO}_3 \cdot \text{A} \cdot \text{MeOH} \cdot \text{A} + \text{B} + \text{MeMgBr} \cdot \text{Dry} \cdot \text{HC}_1 \cdot \text{Et}_2\text{O} > \text{B} \cdot \text{Et}_2\text{O} > \text{C} \cdot \text{C} + \text{NH}_4\text{Cl}(\text{aq}) > \text{D} \cdot 20\% \text{ HCl} > \text{E} \cdot \text{FSI-D-CHM/44} \cdot 4 \cdot \text{PAGE 5} \cdot \text{Ph} > 90\% \cdot \text{O} \cdot \text{Ph} \cdot \text{HO} > \text{Ph} > 90\%$ Write the appropriate reagent necessary for the following two conversions and account for their formation : 10 Write the structure of the product of the following reaction and the corresponding mechanistic pathway : 5 $\text{SnCl}_4 \cdot \text{CO}_2\text{Me}$ Classify the following sigmatropic rearrangement and suggest if it would proceed readily or slowly. Explain your answer : 5 $113\text{C} \cdot \text{CO}_2\text{C}_2\text{H}_5 \rightarrow \text{CN} \cdot \text{H}_3\text{C}, < \text{CO}_2\text{C}_2\text{H}_5$

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49. [UNKNOWN] paper-ii

4. Consider the following statements regarding diagonal relationship of 'B' and 'Si':

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50. [UNKNOWN] paper-ii

4. It is a rare earth metal Which of the statements given above are correct? (a) 1, 2, 3 and 4 (b) 2 and 4 only (c) 1, 2 and 4 only (d) 1 and 3 only

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51. [UNKNOWN] paper-ii

4. O, O₂²⁻ → Both paramagnetic Which of the pair(s) of oxygen species given above represents the correct magnetic behavior? (a) 1 only (b) 1 and 2 (c) 3 (d) 2 and 4

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52. [UNKNOWN] paper-ii

4. NO⁺ has less bond order than NO Which of the statement(s) given above is/are correct? (a) 1 and 4 (b) 1 and 2 (c) 3 (d) 2 only

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53. [UNKNOWN] paper-ii

4. The end point is observed since metal-In complex is more stable than metal-EDTA complex Which of the statement(s) given above is/are correct? (a) 1, 3 and 4 (b) 2, 3 and 4 (c) 2 and 3 only (d) 2 only ===== PAGE 8 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Chemistry.pdf

54. [UNKNOWN] paper-ii

4. Internal energy Which of the following statements given below are correct? (a) 1 and 3 are intensive quantities; 2 and 4 are extensive quantities (b) 1 and 4 are extensive quantities; 2 and 3 are intensive quantities (c) 1 and 2 are extensive quantities; 3 and 4 are intensive quantities (d) 1, 2 and 4 are extensive quantities; 3 is intensive quantity S° F E D C A B T1 T2 Temperature ===== PAGE 11 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Chemistry.pdf

55. [UNKNOWN] paper-ii

Q4. (a) A rigid and insulated tank of volume 2 m³ is divided into two equal compartments by a very thin partition. One compartment contains an ideal gas at 400 K and 3 MPa while the second compartment contains the same gas at 600 K and 1 MPa. The partition is punctured and the gases are allowed to mix. Determine the entropy change of the gas. The isobaric molar heat capacity of the gas is equal to (5/2)R. The value of R may be taken as 8.314 J/mol.K. 15 Estimate the standard Gibbs free energy of formation of CH₃OH (g) from the known value of standard Gibbs free energy of CH₃OH (l) which is given as ΔG°_f = -166.215 kJ. The saturation pressure of CH₃OH at 298 K is 0.16716 bar. 10 A saturated liquid at 1500 kPa and 500 K, with an enthalpy of 750 kJ/kg is throttled to a liquid-vapour mixture at 150 kPa and 300 K. At the exit conditions, the enthalpy of the standard liquid is 500 kJ/kg and the enthalpy of the saturated vapour is 2500 kJ/kg. Find the percentage of the original liquid which vaporises. 15 FSI-D-CHME 3 ===== PAGE 4 ===== SECTION B

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56. [UNKNOWN] paper-ii

4. (a) When 4-bromonitrobenzene is reacted with cyanide ion, the product is devoid of any nitrogen. Write the structure of the product, name the reaction and write the corresponding mechanism. 5 (b) Write the structure of the major product in the following two reactions and justify their formation : 5 EtOe > (c) Carry out the following transformations : 5x2=10 N Glycerol → (ii) HCHO → C(CH₃)₄ FSI-D-CHM/44 5 [P.T.O. ===== PAGE 6 ===== (d) Write the structure of the products (A to E) in each step of the following sequence of reactions : 10 0 CH₃ H₃ AlCl₃ Py/Ac₂O B mCPBA A > > C A Hydrolysis FeCl₃ > D E What happens when salicylic acid is reacted with bromine in acetic acid? Justify your answer. 5 Would you require the following conversion to require heat or light? Explain. 5 SECTION-B

57. [UNKNOWN] paper-ii

5. Consider the following statements regarding Lanthanum:

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58. [UNKNOWN] paper-ii

Q5. Answer all of the following questions: 84=40 What is black liquor ? What are the chemicals present in it? 8 What is plant layout ? How is it prepared? 8 What are the legal aspects of safety in plant design? 8 Explain primary and secondary air pollutants. 8 How is present worth determined ? What is discrete payment present worth factor? 8

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59. [UNKNOWN] paper-ii

5. Answer the following questions : 5x8=40 Calculate M_n of a polymer consisting of three fractions with molecular weights 1×10^5 , 2×10^5 and 3×10^5 . The mole fractions are found to be 0.1, 0.5 and 0.4 respectively. Zimmer plot constructed from light scattering results on polymer solutions, provides useful information. Justify your answer. Determine the molecular weight of a polystyrene sample. Given $a = 0.60$, $K = 1.6 \times 10^{-4}$ dL/g and limiting viscosity number $n = 0.4$ dL/g. Compound A consisting of carbon, hydrogen and oxygen shows in $^1\text{H-NMR}$ (100 MHz) a singlet (9H) in the upfield region, another singlet (2H) in downfield compared to the previous protons and a third broad singlet (1H) that appears in the most downfield region. A on heating with H_2SO_4 produces B, which again on catalytic hydrogenation produces 2-methylbutane. Write the structures of A and B, and account for the $^1\text{H-NMR}$. 4-Methyl-3-penten-2-one has two absorption bands in its UV spectrum—one at 236 nm and one at 314 nm. $\text{CH}_3\text{CH}_2\text{CCH}=\text{C}-\text{CH}_3$ Why are there two absorption bands? Which band shows the greater absorbance? FSI-D-CHM/44 6 ===== PAGE 7 ===== (f) The mass spectra of two different cycloalkanes both show a molecular ion peak at $m/z = 98$. One spectrum shows a base peak at $m/z = 69$ and the other shows a base peak at $m/z = 83$. Identify the cycloalkanes. (g) Give the product of the following reaction : 0 II i) NaBH_4 $\text{CH}_3\text{CH}=\text{CHCH}_2-\text{C}-\text{Cl}$ 3 Predict the stereochemistry of the product in the following reaction : „CH 3 NCH ci CH3 C H3

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60. [UNKNOWN] paper-ii

5. Attempt the following: Illustrate 'McLafferty Rearrangement' with a suitable example. 5 Calculate X of the given compound (A) by the use of Woodward-Fieser rule. (A) 5 0 Write the names and draw the structures of two purins and three pyrimidines that occur in RNA and/or DNA. 5 Which of the following will show a microwave rotational spectrum and why? HCl , CH_4 , H_2O and CH_3Cl 5 The rotational spectrum of CO shows a series of lines placed 3.84235 cm^{-1} apart. Calculate the moment of inertia of $\text{C}=\text{O}$ bond. [$h = 6.626 \times 10^{-27} \text{ erg sec.}$] 5 Show all the possible fundamental modes of vibration of CO_2 molecule. 5 Explain the E.S.R. spectrum of the methyl radical. 5 In ^{11}B NMR spectrum of a compound at 200 MHz, a signal at 1058 Hz was observed. Calculate the chemical shift (δ) for this proton. 5 (i) Arrange $-\text{O}-$, $-\text{C}=\text{O}$ of the given compounds in decreasing order and give suitable explanation of your answer. Et° cy.0 (C) 5 (A) (ii) What are silicones? Mention some of its uses. 5 What is cross-linking in the context of polymers? Suggest one method for obtaining polyvinyl alcohol. 1 0 fsi-d-chn 4 ===== PAGE 5 ===== What is a peptide bond? What is the difference between a peptide, oligopeptide and protein? 1 0 (i) What is borazine ? Why is it called 'Inorganic benzene' ? 5 (ii) What is Zeigler-Natta catalysis? What is the active form of this catalyst? 5 7.(a) (.) Give stereochemistry of the product of the given reaction: 0 A (6) H_3PO_4 h) (A) H_3PO_4 5 (ii) Complete the following reaction: (c) Me IA Me 5 (D) (i) 'In electronic spectra a bathochromic shift is often associated with increasing conjugation' — Explain. 5 (ii) Aniline on treatment with sodium nitrite and HCl in cold generates a product (X) which on treatment with β -naphthol gives (Y). Provide the structures of X and Y. Name the concerned reaction. 5 Write a short note on 'Crossover Experiment' with special emphasis on the elucidation of reaction mechanism. 1 0 Predict the reaction between ethene and allyl cation in the light of Woodward- Hoffmann rules. 10 8.(a) Write a short note on 'Determination of molecular weight by viscosity measurements'. 10 8.(b) Starting from benzene, how will you prepare the following employing Friedel- Crafts reaction? 4-oxo-4-phenylbutanoic acid. Isobutyl benzene. 1 0 8.(c) Complete the given reactions and give its mechanism: 0 $\text{CH}_3\text{h}\nu$ (i) $\text{CH}_3-\text{C}-\text{CH}_2-\text{CH}_2-\text{C}_6\text{H}_5$ Norrish Type - II CH_3 reaction

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61. [UNKNOWN] paper-ii

5 - A 5 fsi-d-chm ===== PAGE 6 ===== 8.(d) Complete the given reaction and give its mechanism : $\text{Ph}_3\text{P Br} - \text{CH}_2 - \text{C}_6\text{H}_5$ / NaH / THF 5 A (ii) Complete the following reaction sequence and give mechanism for the formation of (B) into (C) : SeO_2 (i) cone $\text{NaOH} / \text{A} \text{ Cl} - \text{I} - \text{CHO} \longrightarrow (\text{A}) > (\text{B})$ (ii) H^+ 5 fsi-d-chin 6

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62. [UNKNOWN] paper-ii

6. Argon (Ar) is used in arc welding because of its: (a) High calorific value (b) Ability to lower the melting point of metal (c) Low reactivity (d) Flammability ===== PAGE 2 =====

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63. [UNKNOWN] paper-ii

Q6. Explain the production of sugar with flow diagram. Also give major engineering problems associated with the production. 15 Why is hydrogenation of edible oil required ? Explain hydrogenation of vegetable oil with suitable diagram. 15 Explain the relative advantages and disadvantages of updraft and downdraft biomass gasifier. 10

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64. [UNKNOWN] paper-ii

6. (a) Give reasons for the following : $5 \times 4 = 20$ Teflon is an inert polymer, while LDPE is a branched polymer. Natural rubber on masticating with about 30% sulphur becomes a hard plastic. Proteins are considered as monodispersed polymers. Threshold molecular weight of polyethylene is higher than that for nylons. Ethylbenzene shows a characteristic peak at $m/z = 91$, while n-propylbenzene shows strong peak at $m/z = 92$. Explain with the help of fragmentation pattern. 10 Identify the compounds X and Y in the following transformation : $5 \text{ cyH}_3 \text{ mCPBA LiAlH}_4 \text{ X Y CHCl}_3 \text{ Et}_2\text{O}$ Which will show an oxygen-hydrogen stretch at a greater wave number— ethanol dissolved in carbon disulfide or an undiluted sample of ethanol? Justify your answer briefly. 5

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65. [UNKNOWN] paper-ii

7. Which one of the following represents the correct order of ionic radii among K^+ , S^{2-} , Cl^- and Ca^{2+} ? (a) $\text{Ca}^{2+} > \text{K}^+ > \text{Cl}^- > \text{S}^{2-}$ (b) $\text{S}^{2-} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$ (c) $\text{Cl}^- > \text{S}^{2-} > \text{K}^+ > \text{Ca}^{2+}$ (d) $\text{K}^+ > \text{Ca}^{2+} > \text{S}^{2-} > \text{Cl}^-$

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66. [UNKNOWN] paper-ii

Q7. Discuss various methods for the extinction of fire. Differentiate between Hazard and Risk. 15 Discuss the various plume patterns for the different environmental prevailing lapse rate. 10 What are biodegradable and non-biodegradable organics ? Explain in short, various methods used to treat biodegradable organics present in the waste water. 15 1 F5I-D-CHME 4 ===== PAGE 5 =====

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67. [UNKNOWN] paper-ii

7. (a) Write the structure of the major and minor products formed in the following reaction and give a suitable explanation : $10 \text{ Na, NH}_3(1) > \text{ii) H}_2\text{O}$ 7 FSI-D-CHM/ 44 [P.T.O. ===== PAGE 8 ===== (b) Justify your answers for the following statements : $5 \times 4 = 20$ Light scattering is a highly useful technique for the determination of molecular weight of a polymer. The determination of molecular weight by end-group analysis of a polymer is carried out only if the number of determinable groups of molecules is known. Osmometric method is preferred over viscosity method for the determination of molecular weight of a polymer. Softening temperature of nylon 6,6 is higher than that of nylon 6,10. (c) Of the following two compounds A and B, one cannot be oxidized by periodic acid. Identify it and justify your answer : 10 OH A

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68. [UNKNOWN] paper-ii

8. The dipole moments of $\text{H} - \text{C} \equiv \text{C} - \text{Cl}$, CO and HCl are in the directions: (a) $\leftarrow$, $\leftarrow$ and $\leftarrow$ respectively (b) $\leftarrow$, $\leftarrow$ and $\rightarrow$

respectively (c) $\rightarrow$, $\rightarrow$ and $\rightarrow$ respectively (d) $\leftarrow$, $\rightarrow$ and $\leftarrow$ respectively

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69. [UNKNOWN] paper-ii

Q8. (a) An equipment has an initial cost of 12,000. The depreciation will be charged as by making equal charges each year and first payment being made at the end of the first year. Determine the yearly cost due to depreciation. Useful life period of equipment is seven years. Salvage value is 1000 and depreciation fund will be accumulated at an annual interest rate of 4.5%. 15 The purchase cost of a heat exchanger of 20 m² area was 5,00,000 in 2007. What will be the estimated cost of a similar heat exchanger of 50 m² area in the year 2014 ? Assume the six-tenth factor rule for scaling and the cost index for 2007 as 430.2. The projected cost index for the year 2014 is 512.6. 10 The total capital investment for a chemical plant is 20,00,000 and its working capital is 2,00,000. If the plant produces an average of 10,000 kg final products per day during 365 days/year, what selling price in per kg of product would be necessary to give a turnover ratio of 1? 15 F51-0-CHME 5

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70. [UNKNOWN] paper-ii

8. (a) 2,2-Dimethylcyclopropanone undergoes ring-opening when attacked by methoxide ion. The product (b.p. 101 °C) has the following spectral properties: IR : 1740 cm⁻¹ (s), 1160 cm⁻¹ (no absorption near 1600 cm⁻¹ or 3100 cm⁻¹) 1H-NMR : δ 3.6 (3H, singlet), 81.2 (9H, singlet) UV : Transparent above 200 nm Mass : m/z 116, 85, 59, 31 Deduce the structure of the product and assign the spectral data. From the mechanism, an alternative product might have arisen. What is it? 15 (b) (i) Write the structure of the product in the following reaction : Ph HO BH₃-THF Ph

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71. [UNKNOWN] paper-ii

9. Consider the following pairs of species of oxygen and their magnetic behavior:

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72. [UNKNOWN] paper-ii

10. Consider the following statements regarding the properties of Nitric oxide (NO):

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73. [UNKNOWN] paper-ii

11. Which one of the following substances has the greatest ionic character? (a) Cl₂O (b) NCl₃ (c) PbCl₂ (d) BaCl₂
===== PAGE 3 =====

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74. [UNKNOWN] paper-ii

12. Consider the following statements regarding Madelung constant: Statement-1: Madelung constant typically increases with coordination number Statement-2: Large contribution towards Madelung constant comes from nearest neighbors, and such neighbors are more numerous when coordination number is large Which one of the following is correct in respect of above statements? (a) Both statement-1 and statement-2 are true and statement-2 is the correct explanation of statement-1 (b) Both statement-1 and statement-2 are true but statement-2 is not the correct explanation of statement-1 (c) Statement-1 is true but statement-2 is false (d) Statement-1 is false but statement-2 is true

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75. [UNKNOWN] paper-ii

13. Which one of the following represents the correct order of melting point among BeCl₂, MgCl₂, CaCl₂ and NaCl? (a) BeCl₂ < MgCl₂ < CaCl₂ < NaCl (b) MgCl₂ < BeCl₂ < CaCl₂ < NaCl (c) BeCl₂ < MgCl₂ < NaCl < CaCl₂ (d) BeCl₂ < CaCl₂ < MgCl₂ < NaCl

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76. [UNKNOWN] paper-ii

14. What kind of defect(s) appear in AgBr crystal? (a) Schottky and Frenkel (b) Frenkel only (c) Schottky only (d) Metal excess defect

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77. [UNKNOWN] paper-ii

14- THF, Room temp. Me How many moles of BH₃ are needed to react with 2 moles of the starting material? 5 What stereoisomer would be formed from the reaction of cyclopentene with OsO₄ followed by hydrolysis? Explain. 5 Draw the ESR spectrum of CH₃CHOCH₂CH₃. 5 (c) Sketch the chemical reactions in the preparation of polystyrene starting from benzene and list a few important properties. 10 * * * FSI-D-CHM/44 8 BS9-90

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78. [UNKNOWN] paper-ii

16. The correct order of the bond angles among NH₃, NH₄⁺ and NH₂⁻ is: (a) NH₃ > NH₄⁺ > NH₂⁻ (b) NH₄⁺ > NH₃ > NH₂⁻ (c) NH₃ > NH₂⁻ > NH₄⁺ (d) NH₄⁺ > NH₂⁻ > NH₃ ===== PAGE 4 =====

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79. [UNKNOWN] paper-ii

17. Consider the following statements: Statement-1: KO₂ is paramagnetic Statement-2: The super oxide O₂⁻ is paramagnetic because it has one unpaired electron in π^* 2p molecular orbital Which one of the following is correct in respect of above statements? (a) Both statement-1 and statement-2 are true and statement-2 is the correct explanation of statement-1 (b) Both statement-1 and statement-2 are true but statement-2 is not the correct explanation of statement-1 (c) Statement-1 is true but statement-2 is false (d) Statement-1 is false but statement-2 is true

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80. [UNKNOWN] paper-ii

18. Which one of the following is NOT a resonance structure of cyanate (OCN⁻)?

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81. [UNKNOWN] paper-ii

19. What are the spin only magnetic moments (in BM) for Ni (II) ion in a square planar and octahedral geometry, respectively? (a) 0 (zero) and 2.83 (b) 2.83 and 0 (zero) (c) 2.83 and 2.83 (d) 0 (zero) and 0 (zero)

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82. [UNKNOWN] paper-ii

20. Which one of the following complexes is tetrahedral as well as diamagnetic? (a) [Ni(NH₃)₄]²⁺ (b) [NiCl₄]²⁻ (c) [Ni(CO)₄] (d) [Ni(CN)₄]²⁻

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83. [UNKNOWN] paper-ii

21. Which one of the following will NOT precipitate AgCl while reacting with excess of AgNO₃ solution at room temperature? (The oxidation state of cobalt (Co) in all the complexes is +3) (a) CoCl₃.6NH₃ (b) CoCl₃.5NH₃ (c) CoCl₃.4NH₃ (d) CoCl₃.3NH₃ ===== PAGE 5 =====

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84. [UNKNOWN] paper-ii

22. Which one of the following complexes CANNOT be ionized in solution? (a) [Pt(NH₃)₆]Cl₄ (b) K₂[PtF₆] (c) K₄[Fe(CN)₆] (d) [Co(NH₃)₃Cl₃]

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85. [UNKNOWN] paper-ii

23. The maximum denticity of the polydentate ligand tetraethylene pentamine is equal to: (a) 4 (b) 2 (c) 5 (d) 3

86. [UNKNOWN] paper-ii

24. The correct IUPAC name of $K_2[Zn(OH)_4]$ is: (a) Dipotassium tetrahydroxo zinc (II) (b) Potassium tetrahydroxo zinc (II) (c) Potassium tetrahydroxido zincate (II) (d) Dipotassium tetrahydroxido zincate (II)

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87. [UNKNOWN] paper-ii

25. Which one of the following octahedral complexes does NOT show geometrical isomerism? ('A' and 'B' are monodentate ligands) (a) $[MA_2B_4]$ (b) $[MA_3B_3]$ (c) $[MA_4B_2]$ (d) $[MA_5B]$

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88. [UNKNOWN] paper-ii

26. Which one of the following will NOT show optical isomerism? (a) $[Cr(C_2O_4)_3]^{3-}$ (b) $Cis-[PtCl_2(en)_2]^{2+}$ (c) $[Co(EDTA)]^-$ (d) $[PtCl_2(NH_3)_2]$

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89. [UNKNOWN] paper-ii

27. How many isomers are possible for $[Co(en)_2Cl_2]^+$ complex ion? (a) 1 (b) 2 (c) 3 (d) 4

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90. [UNKNOWN] paper-ii

28. Which one of the following equilibria will shift to the left in response to increase in pressure? (a) $N_2 + 3H_2 \rightleftharpoons 2NH_3 + 22.2 \text{ Kcal}$ (b) $H_2 + I_2 \rightleftharpoons 2HI + 12.4 \text{ Kcal}$ (c) $N_2 + O_2 \rightleftharpoons 2NO - 43.2 \text{ Kcal}$ (d) $N_2O_4 \rightleftharpoons 2NO_2 + 14 \text{ Kcal}$ ===== PAGE 6 =====

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91. [UNKNOWN] paper-ii

29. The correct order of the acidic strength of H_3PO_4 , H_3PO_3 and H_3PO_2 is: (a) $H_3PO_2 > H_3PO_3 > H_3PO_4$ (b) $H_3PO_4 > H_3PO_3 > H_3PO_2$ (c) $H_3PO_4 > H_3PO_2 > H_3PO_3$ (d) $H_3PO_3 > H_3PO_4 > H_3PO_2$

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92. [UNKNOWN] paper-ii

30. Which one of the following substances has highest proton affinity? (a) H_2O (b) NH_3 (c) H_2S (d) CH_3COOH

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93. [UNKNOWN] paper-ii

31. An aqueous solution of HN_3 was diluted two times, three times, four times and five times using required amount of water. Considering ' α ' to be the ionization constant and ' V ' as the final volume of diluted solution, choose the correct answer from the following: (a) All the solutions will be acidic and the graph of α vs. V is non-linear (b) All the solutions will be basic and the graph of α vs. V is non-linear (c) All the solutions will be acidic and the graph of α vs. V is linear (d) All the solutions will be basic and the graph of α vs. V is linear

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94. [UNKNOWN] paper-ii

32. Which one of the following represents the correct order of acidity of water and alcohols? (a) $H_2O > RCH_2OH > R_2CHOH > R_3COH$ (b) $RCH_2OH > R_2CHOH > R_3COH > H_2O$ (c) $H_2O < RCH_2OH < R_2CHOH < R_3COH$ (d) $RCH_2OH < R_2CHOH < R_3COH < H_2O$

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95. [UNKNOWN] paper-ii

33. A student calculated the pH of 10^{-7} M NaOH solution and 10^{-8} M HCl solution to be 7 and 8, respectively. To verify,

the student performed experiment in laboratory using pH-meter. The instrument displays the pH of former and later solutions as 7.3 and 6.9, respectively. Choose the correct answer from the following: (a) The pH-meter is malfunctioned and the calculation results of the student are correct (b) The pH-meter is working properly and the calculation results of the student are incorrect (c) The pH-meter is malfunctioned and the calculation results of the student are wrong (d) The calculation results of the student are correct and the instrument is working properly only for acids

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96. [UNKNOWN] paper-ii

34. Which one of the following has the pH value NOT equal to one? (a) 0.1 M HNO₃ (b) 0.05 M H₂SO₄ (c) 0.1 M CH₃COOH (d) 50 ml of 0.4 M HCl + 50 ml of 0.2 M NaOH ===== PAGE 7 =====

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97. [UNKNOWN] paper-ii

35. What will be the pH of a buffer solution containing 0.20 mole of NH₄OH and 0.25 mole of NH₄Cl per litre? (Dissociation constant of NH₄OH at room temperature is 1.81×10^{-5} , $\log 1.81 = 0.2577$ and $\log 1.25 = 0.0969$) (a) 9.161 (b) 9.521 (c) 8.161 (d) 9.312

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98. [UNKNOWN] paper-ii

36. The molarity of concentrated commercial HCl with 36 % (w/w) HCl and density 1.19 g/cm³ is close to: (a) 11.7 M (b) 10.6 M (c) 11.2 M (d) 11.5 M

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99. [UNKNOWN] paper-ii

37. Which one of the following primary standard is usually used to standardize KOH by titration? (a) Potassium acid phthalate (b) Sulphuric acid (c) Hydrochloric acid (d) Nitric acid

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100. [UNKNOWN] paper-ii

38. While performing the titration between Mohr's salt solution and potassium dichromate solution using diphenylamine as indicator, a student forgot to add o-phosphoric acid. Choose the correct answer from the following: (a) The end point will be observed before the actual end point (b) The end point will be observed after the actual end point (c) The end point will never be observed (d) The end point will be observed at its actual end point as there was no need to add o-phosphoric acid

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101. [UNKNOWN] paper-ii

39. Consider following statements in respect of titration between Ca²⁺ solution and EDTA solution using Eriochrome Black T as indicator (In):

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102. [UNKNOWN] paper-ii

40. In acid base titration the color change of an indicator is expressed by: $\text{pH} = \text{pK}_{\text{In}}' + \log \frac{[\text{InB}]}{[\text{InA}]}$ (Where InB = Indicator in the basic form, InA = indicator in acid form) The basic color of the indicator would be visible when: (a) $\frac{[\text{InB}]}{[\text{InA}]} > 10$ (b) $\frac{[\text{InB}]}{[\text{InA}]} < 10$ (c) $\frac{[\text{InA}]}{[\text{InB}]} > 10$ (d) $[\text{InB}] = [\text{InA}]$

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103. [UNKNOWN] paper-ii

41. Which one of the following comparisons of the average translational kinetic energy (K.E.) and the average molecular speeds of H₂ and N₂ gases at 300 K is correct? Average translational K.E. Average molecular speed (a) H₂ = N₂ H₂ = N₂ (b) H₂ < N₂ H₂ > N₂ (c) H₂ = N₂ H₂ < N₂ (d) H₂ = N₂ H₂ > N₂

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104. [UNKNOWN] paper-ii

42. The expression of isothermal compressibility factor is: (a) $\beta = \frac{1}{P} \left(\frac{\partial V}{\partial P} \right)_T$ (b) $\beta = -\frac{1}{P} \left(\frac{\partial V}{\partial P} \right)_T$ (c) $\beta = \frac{1}{P} \left(\frac{\partial V}{\partial T} \right)_P$ (d) $\beta = -\frac{1}{P} \left(\frac{\partial V}{\partial T} \right)_P$

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105. [UNKNOWN] paper-ii

43. An equimolar mixture of H₂ and N₂ is allowed to effuse through a hole. What is the ratio of rate of diffusion of H₂ to that of N₂? (M.W. of H₂ = 2 g mol⁻¹ and M.W. of N₂ = 28 g mol⁻¹) (a) 1 : 3.7 (b) 3.7 : 1 (c) 1 : 14 (d) 14 : 1

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106. [UNKNOWN] paper-ii

44. The temperature at which the average velocity of Nitrogen (N₂) is equal to that of Oxygen (O₂) at 320 K is: (M.W. of N₂ = 28 g mol⁻¹ and M.W. of O₂ = 32 g mol⁻¹) (a) 365.68 K (b) 280 K (c) 225 K (d) 300 K ===== PAGE 9 =====

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107. [UNKNOWN] paper-ii

45. Which one of the following relations for three types of velocities is true for oxygen gas at 300 K? (where $\langle c^2 \rangle^{1/2}$ = root mean square velocity; $\langle c \rangle$ = average velocity; c_p = most probable velocity) (a) $\langle c^2 \rangle^{1/2} > c_p > \langle c \rangle$ (b) $\langle c \rangle > c_p > \langle c^2 \rangle^{1/2}$ (c) $\langle c^2 \rangle^{1/2} > \langle c \rangle > c_p$ (d) $\langle c \rangle > \langle c^2 \rangle^{1/2} > c_p$

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108. [UNKNOWN] paper-ii

46. Consider the following statements regarding the most probable speed of gas particles:

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109. [UNKNOWN] paper-ii

47. One hundred molecules of Oxygen (O₂) escape through a small hole on a container at a pressure of 1 torr and temperature of 300 K per second to vacuum. The number of hydrogen molecules that would escape per second through the same hole under the same conditions of temperature and pressure is: (M.W. of O₂ = 32 g mol⁻¹ and M.W. of H₂ = 2 g mol⁻¹) (a) 1200 (b) 800 (c) 1600 (d) 400

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110. [UNKNOWN] paper-ii

48. The mean free path of a gas is: (a) Independent of temperature but dependent on pressure (b) Independent of pressure but dependent on temperature (c) Independent of both pressure and temperature (d) Dependent on both pressure and temperature

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111. [UNKNOWN] paper-ii

49. Which one of the following is NOT a Maxwell thermodynamic relation? (a) $\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V$ (b) $\left(\frac{\partial S}{\partial P} \right)_T = \left(\frac{\partial V}{\partial T} \right)_P$ (c) $\left(\frac{\partial V}{\partial S} \right)_T = \left(\frac{\partial T}{\partial P} \right)_S$ (d) $-\left(\frac{\partial T}{\partial P} \right)_S = \left(\frac{\partial V}{\partial S} \right)_T$ ===== PAGE 10 =====

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112. [UNKNOWN] paper-ii

50. Consider the following first order elementary reaction: The half-life period of the forward reaction is 10 min. and that of the backward reaction is 1 min. What is the value of equilibrium constant for the above reaction? (a) 0.693 (b) 0.0693 (c) 0.10 (d) 10.0

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113. [UNKNOWN] paper-ii

51. Consider the following figure representing the increase in entropy of a substance from absolute zero to its gaseous

state with temperature change: ΔS° (fusion) and ΔS° (vaporization) are respectively indicated by: (a) AB and BC (b) BC and CD (c) BC and DE (d) CD and DE

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114. [UNKNOWN] paper-ii

53. Joule Thomson coefficient for a gas is given by: (a) $\mu_{JT} = - \frac{1}{C_p} \left(\frac{\partial H}{\partial T} \right)_P$ (b) $\mu_{JT} = - \left(\frac{\partial H}{\partial T} \right)_P$ (c) $\mu_{JT} = - \frac{1}{C_p} \left(\frac{\partial H}{\partial T} \right)_P$ (d) $\mu_{JT} = 1 \left(\frac{\partial H}{\partial T} \right)_P$

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115. [UNKNOWN] paper-ii

54. Consider the following statements regarding Joule Thomson effect:

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116. [UNKNOWN] paper-ii

55. For the reaction $\text{H}_2\text{F}_2(\text{g}) \rightarrow \text{H}_2(\text{g}) + \text{F}_2(\text{g})$, ΔH is equal to: (a) $\Delta E + 2RT$ (b) $\Delta E - 2RT$ (c) ΔE (d) $\Delta E + RT$

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117. [UNKNOWN] paper-ii

56. Which of the following thermodynamic relations are correct?

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118. [UNKNOWN] paper-ii

57. What is the sign of ΔG° and the value of K for an electrochemical cell for which $E^\circ_{\text{cell}} = 0.80\text{V}$? (a) ΔG° is negative and $K > 1$ (b) ΔG° is positive and $K > 1$ (c) ΔG° is positive and $K < 1$ (d) ΔG° is negative and $K < 1$ ===== PAGE 12 =====

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119. [UNKNOWN] paper-ii

58. If $\Delta G = \Delta H - T\Delta S$ and $\Delta G = \Delta H + T \left(\frac{\partial \Delta H}{\partial T} \right)_P$ then the variation of emf of cell (E°) with temperature T at constant pressure is given by: (a) $\frac{\partial E^\circ}{\partial T} = \frac{\Delta S}{nF}$ (b) $\frac{\partial E^\circ}{\partial T} = \frac{\Delta H}{nF}$ (c) $\frac{\partial E^\circ}{\partial T} = \frac{\Delta G}{nF}$ (d) $\frac{\partial E^\circ}{\partial T} = \frac{\Delta H}{nF}$

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120. [UNKNOWN] paper-ii

59. Which of the following is correct about the sign of enthalpy and entropy changes of an evaporating liquid? (a) ΔH is positive; ΔS is positive (b) ΔH is negative; ΔS is negative (c) ΔH is positive; ΔS is negative (d) ΔH is negative; ΔS is positive

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121. [UNKNOWN] paper-ii

60. Consider the following equilibrium process at 373K and 1 atm. pressure: Which one of the following relations is true for the above process? (a) $\Delta H = 0$ (b) $\Delta E = 0$ (c) $\Delta H = T\Delta S$ (d) $\Delta H = \Delta E$

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122. [UNKNOWN] paper-ii

61. Which one of the following aqueous solutions has the lowest vapor pressure at room temperature? (a) 0.1 molal hexaamminecobalt (III) chloride (b) 0.1 molal barium chloride (c) 0.1 molal glucose (d) 0.1 molal Tris (ethylenediamine) copper (II) sulfate

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123. [UNKNOWN] paper-ii

62. Compared to pure water, 1 M sugar-water solution will have a: (a) Lower vapor pressure, lower boiling point and lower freezing point (b) Lower vapor pressure, lower boiling point and higher freezing point (c) Lower vapor pressure, higher boiling point and lower freezing point (d) Higher vapor pressure, higher boiling point and higher freezing point

===== PAGE 13 =====

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124. [UNKNOWN] paper-ii

63. The relation between osmotic pressure (Π) and lowering of vapor pressure in an infinitely dilute solution is: (a) $\Pi = V_0 RT \ln (\frac{p_0}{p})$ (b) $\Pi = \frac{p_0}{V_0} \ln (\frac{p_0}{p})$ (c) $\Pi = V_0 + RT \ln (\frac{p_0}{p})$ (d) $\Pi = V_0 - RT \ln (\frac{p_0}{p})$

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125. [UNKNOWN] paper-ii

64. The vapor pressure of ethyl acetate at 293K is 72.8 torr. The vapor pressure of the solution obtained by dissolving 0.4 g of a non-volatile solute (molar mass = 60 gmol⁻¹) in 100 g of ethyl acetate at the same temperature is: (a) 72.91 torr (b) 72.37 torr (c) 72.60 torr (d) 72.00 torr

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126. [UNKNOWN] paper-ii

65. People with high blood pressure are advised to have lower intake of salt in their food because:

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127. [UNKNOWN] paper-ii

66. The equation $\Pi = c_2 RT$ is known as: (where $c_2 = n_2/V$ is the molar concentration of the solute in the given solution and other symbols have their usual meanings) (a) Van't Hoff equation (b) Raoult's law equation (c) Henry law equation (d) Walden's rule

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128. [UNKNOWN] paper-ii

67. The boiling point of Chloroform was raised by 0.325K when 5.141×10^{-4} kg of a non-volatile solute was dissolved in 35×10^{-3} kg of Chloroform. Calculate the molar mass of the solute? (molar boiling point elevation constant for chloroform is 3.9 K kg mol⁻¹) (a) 0.215 kg mol⁻¹ (b) 0.176 kg mol⁻¹ (c) 0.195 kg mol⁻¹ (d) 0.151 kg mol⁻¹

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129. [UNKNOWN] paper-ii

68. P is the total vapor pressure of an ideal binary liquid solution. If y_A and y_B are the amount fractions of the constituents in vapor phase, then which one of the following statements is correct? (a) P varies linearly with y_A and y_B (b) $1/P$ varies linearly with y_A and y_B (c) P varies linearly with y_A and y_B and passes through origin (d) $1/P$ varies linearly with y_A and y_B and passes through origin

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130. [UNKNOWN] paper-ii

69. The Henry Law constant (K_H) for Hydrogen (H_2) in water at room temperature is 7.03×10^9 Pa. If the pressure of H_2 in equilibrium with water is 1.00×10^6 Pa at the same temperature, then the mole fraction of H_2 in the solution is: (a) 0.00427 (b) 0.0427 (c) 0.000821 (d) 0.000142

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131. [UNKNOWN] paper-ii

70. Specific conductance of a decimolar solution of KCl at 18°C is 1.12 Sm⁻¹. The resistance of a conductivity cell containing the solution at the same temperature was found to be 55 ohm. What is the cell constant? (a) 0.616 cm⁻¹ (b) 0.716 cm⁻¹ (c) 0.516 cm⁻¹ (d) 0.699 cm⁻¹

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132. [UNKNOWN] paper-ii

71. In acidic medium MnO_4^- – converted to Mn^{2+} . The quantity of electricity in Faraday required to reduce 0.2 mole of

MnO₄ – to Mn²⁺ would be: (a) 0.2 (b) 0.4 (c) 1 (d) 2.5

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133. [UNKNOWN] paper-ii

72. What will be the correct order of mobility of the following ions in aqueous solution at 298 K? (a) Cl⁻ > F⁻ > SO₄²⁻ > Br⁻ (b) F⁻ > SO₄²⁻ > Br⁻ > Cl⁻ (c) SO₄²⁻ > Br⁻ > Cl⁻ > F⁻ (d) Br⁻ > SO₄²⁻ > Cl⁻ > F⁻

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134. [UNKNOWN] paper-ii

73. At 291 K the ionic velocities of Ag⁺ is 0.00057 cm/s and that of NO₃⁻ is 0.00063 cm/s. If the specific conductivity of 0.1N AgNO₃ solution is 0.00947 ohm⁻¹cm⁻¹, then the degree of dissociation at this dilution is: (a) 0.817 (b) 0.918 (c) 0.698 (d) 0.761 ===== PAGE 15 =====

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135. [UNKNOWN] paper-ii

74. Plots of equivalent conductance (Λ) vs square root of concentration ($\sqrt{C}$) of two electrolytes are given below, with Λ_0 being the equivalent conductance at infinite dilution: (I) (II) Which one of the following statements is correct? (a) Plot I is for weak electrolyte and plot II is for strong electrolyte (b) Extrapolation to Λ_0 as shown in plot I cannot be done (c) Extrapolation to Λ_0 in plot II cannot be done to get the correct Λ_0 (d) Plots I and II are valid for any electrolyte for different range of concentrations

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136. [UNKNOWN] paper-ii

75. Consider the following statements for an electrolyte solution which is diluted to approach infinite dilution: Statement-I: Ion-ion interactions are reduced to effect the increase in conductance Statement-II: The number of ions gets increased to effect increase in conductance Which one of the following is correct with respect to above statements? (a) Both the statements I and II are true for strong electrolytes (b) Both the statements I and II are true for weak electrolytes (c) Statement I is true for weak electrolytes and statement II is true for strong electrolytes (d) Statement I is true for strong electrolytes and statement II is true for weak electrolytes

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137. [UNKNOWN] paper-ii

76. The value of $\log \gamma_{\pm}$ in the Debye-Huckel limiting law for a 0.01m KCl solution in water at 25 °C is: (where $\gamma_{\pm}$ is the mean ionic activity coefficient) (a) - 0.1018 (b) - 0.0509 (c) + 0.1018 (d) - 0.0254 $\wedge \sqrt{C} \wedge \Lambda_0 \sqrt{C}$ ===== PAGE 16 =====

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138. [UNKNOWN] paper-ii

77. Consider the following plot: The above plot indicates the conductometric titration between: (a) Weak acid vs weak base (b) Strong acid vs weak base (c) Weak acid vs strong base (d) Strong acid vs strong base

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139. [UNKNOWN] paper-ii

78. Consider the following plot for conductometric study of a reaction between KCl and AgNO₃ solutions: Cause/s behind the constancy of conductance in the region PQ is/are due to: (a) Precipitation of AgCl only (b) Replacement of Cl⁻ by NO₃⁻ with each species having equal conductance (c) Precipitation of AgCl and replacement of Cl⁻ by NO₃⁻ (d) Inert behavior of the instrument in this region Conductance P Q Volume of AgNO₃ solution R Conductivity Volume of base ===== PAGE 17 =====

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140. [UNKNOWN] paper-ii

79. Consider the following plot for the conductometric titration: Which one of the following is correct in respect of above plot? (a) Plot I is for weak acid vs weak alkali; plot II is for weak acid vs strong alkali (b) Plot I is for strong acid vs strong

alkali; plot II is for AgNO₃ solution vs NH₄Cl solution (c) Plot I is for AgNO₃ solution vs HCl; plot II is for AgNO₃ vs KCl (d) Plot I is for weak acid vs strong alkali; plot II is for weak acid vs weak alkali

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141. [UNKNOWN] paper-ii

80. Two electrolytes AX and BX were chosen to determine transport number using moving boundary method. It is given that: I: The cation B⁺ moves slower than the cation A⁺ II: The electrolyte BX is denser than AX Which one of the following is correct? (a) The transport number of A⁺ can be determined if AX is used as principal electrolyte and BX as indicator electrolyte (b) The transport number of B⁺ can be determined if AX is used as indicator electrolyte and BX as principal electrolyte (c) The transport number of X⁻ can be determined if AX is used as indicator electrolyte and BX as principal electrolyte (d) The transport number of A⁺, B⁺ and X⁻ can be determined if AX is used as principal electrolyte and BX as indicator electrolyte

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142. [UNKNOWN] paper-ii

81. The total number of delocalized pi-electrons in compounds I and II are: (a) 18 and 14 respectively (b) 24 and 16 respectively (c) 16 and 12 respectively (d) 22 and 14 respectively Volume of titrant Conductance I II ===== PAGE 18 =====

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143. [UNKNOWN] paper-ii

82. Rank the following structures in order of increasing contribution to the resonance hybrid: (a) II < I < III (b) III < II < I (c) III < I < II (d) I < II < III

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144. [UNKNOWN] paper-ii

83. Which of the following indicated atoms in each species has sp²-hybridization? (a) I and III only (b) I, III and IV only (c) II and IV only (d) I, II, III and IV

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145. [UNKNOWN] paper-ii

84. The total number of hyperconjugative structures possible for the following carbocation is: (a) 3 (b) 1 (c) 4 (d) 12

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146. [UNKNOWN] paper-ii

85. Which one of the following statements regarding polarity of bonds is NOT true? (a) A compound with only one polar bond must be polar (b) A compound with two or more polar bonds will always be polar (c) A compound cannot be polar if it contains only nonpolar bonds (d) More electronegative atoms "pull" electron density towards them, making a dipole ===== PAGE 19 =====

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147. [UNKNOWN] paper-ii

86. The correct order of the basicity of the following amines is: (a) IV > III > II > I (b) III > IV > I > II (c) IV > III > I > II (d) III > IV > II > I

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148. [UNKNOWN] paper-ii

87. The correct order of stability of the following carbocations is: (a) II > IV > I > III (b) I > II > III > IV (c) II > I > IV > III (d) I > III > II > IV

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149. [UNKNOWN] paper-ii

88. Rank the following compounds in order of their decreasing dipole moment: (a) III > II > IV > I (b) IV > III > I > II (c) II > IV > III > I (d) I > II > IV > III

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150. [UNKNOWN] paper-ii

89. The relationship between the following pair of compounds is: (a) Identical (b) Enantiomers (c) Diastereomers (d) Constitutional isomers ===== PAGE 20 =====

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151. [UNKNOWN] paper-ii

90. Consider the following set of compounds: Which one of the following statements is NOT true about the above compounds (I–IV)? (a) I and II represent same structures and are meso compounds (b) [I and III] and [I and IV] are diastereomers (c) III and IV are enantiomers (d) III and IV are diastereomers

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152. [UNKNOWN] paper-ii

91. The correct statement about the following molecule is: (a) Molecule is chiral and possesses a chiral plane (b) Molecule is chiral and possesses a chiral axis (c) Molecule is achiral and possesses a plane of symmetry (d) Molecule is achiral and possesses a centre of symmetry

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153. [UNKNOWN] paper-ii

92. A sample of lactic acid was found to have an enantiomeric excess (ee) of 14% with respect to R (+) isomer. What is the percentage of S (–) isomer in the sample? (a) 86 (b) 57 (c) 43 (d) 72

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154. [UNKNOWN] paper-ii

93. In 2-butanol observed angle of rotation is 1.03° , concentration of sample is 1M and length of sample tube is 10 cm. Specific angle of rotation will be: (a) 1.392° (b) 13.92° (c) 14.20° (d) 139.2°

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155. [UNKNOWN] paper-ii

94. The absolute configuration of C2 and C3 in the following compound is: (a) 2R, 3S (b) 2S, 3R (c) 2S, 3S (d) 2R, 3R ===== PAGE 21 =====

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156. [UNKNOWN] paper-ii

95. The configuration of C1 and C2 carbons in the following molecule is: (a) 1S, 2R (b) 1R, 2R (c) 1R, 2S (d) 1S, 2S

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157. [UNKNOWN] paper-ii

96. The most stable conformation of the following compound is: (a) (b) (c) (d)

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158. [UNKNOWN] paper-ii

97. The stereochemical notations for the following compound is: (a) 2E, 4E, 6S (b) 2Z, 4E, 6S (c) 2Z, 4E, 6R (d) 2Z, 4Z, 6R

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159. [UNKNOWN] paper-ii

98. The total number of geometrical isomers for the following compound is: (a) 4 (b) 5 (c) 6 (d) 7 ===== PAGE 22 =====

160. [UNKNOWN] paper-ii

99. The correct order of stability of Newman projections of 1, 2-dichloroethane is: (a) anti > eclipsed > gauche (b) gauche > eclipsed > anti (c) gauche > anti > eclipsed (d) anti > gauche > eclipsed

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161. [UNKNOWN] paper-ii

100. Among the following, the Newman projections for meso-2,3-butanediol are: (a) I and II (b) I and III (c) III and IV (d) II and IV

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162. [UNKNOWN] paper-ii

101. The major products [X] and [Y] formed in the following reactions are: (a) (b) (c) (d)

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163. [UNKNOWN] paper-ii

102. Which one of the following chlorides undergoes a fastest SN2 reaction with KI in acetone? (a) (b) (c) (d) =====
PAGE 23 =====

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164. [UNKNOWN] paper-ii

103. Indicate the order of reactivity of HX in the following reaction: (a) HI > HBr > HCl (b) HCl > HBr > HI (c) HBr > HI > HCl (d) HBr > HCl > HI

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165. [UNKNOWN] paper-ii

104. Which isomer of 1,2,3,4,5,6-hexachlorocyclohexane shown below undergoes E2 elimination (dehydrohalogenation) most slowly? (a) (b) (c) (d)

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166. [UNKNOWN] paper-ii

105. The following reaction is an example of: (a) E2 elimination (b) E1 elimination (c) E1cb elimination (d) Syn elimination =====
PAGE 24 =====

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167. [UNKNOWN] paper-ii

106. Which of the following statement is TRUE for the reaction mentioned below? (a) I is major and it is called Hoffmann product (b) I is minor and it is called Zaitsev product (c) II is major and it is called Zaitsev product (d) II is minor and it is called Hoffmann product

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168. [UNKNOWN] paper-ii

107. Which one of the following alcohols dehydrates the fastest when heated with acid?

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169. [UNKNOWN] paper-ii

108. The major product formed in the following reaction is: (a) (b) (c) (d) =====
PAGE 25 =====

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170. [UNKNOWN] paper-ii

109. Which one of the following alkenes will react most rapidly with concentrated sulphuric acid? (a) H₂C=CH₂ (b)

(CH₃)₂C=CH₂ (c) Cl₂C=CCl₂ (d) F₃CCH=CHCH₃

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171. [UNKNOWN] paper-ii

110. The regioselective electrophilic addition of HOCl with propene proceeds through: (a) (b) (c) (d)

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172. [UNKNOWN] paper-ii

111. Which one of the following products is formed when bromine adds to anthracene? (a) (b) (c) (d)

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173. [UNKNOWN] paper-ii

112. The correct sequence of reactions to accomplish the following transformation is: (a) (i) CH₃CH₂Cl/AlCl₃, (ii) Cl₂/FeCl₃ (b) (i) Cl₂/FeCl₃, (ii) CH₃CH₂Cl/AlCl₃ (c) (i) CH₃COCl/AlCl₃, (ii) Cl₂/FeCl₃, (iii) Zn-Hg/HCl (d) (i) CH₃COCl/AlCl₃, (ii) Zn-Hg/HCl, (iii) Cl₂/FeCl₃ ===== PAGE 26 =====

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174. [UNKNOWN] paper-ii

113. The intermediate [X] and the major product [Y] formed in the following reaction are: (a) (b) (c) (d)

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175. [UNKNOWN] paper-ii

114. Which is the major product formed in the reaction of phenol with bromine in carbon disulfide at 0– 5°C temperature? ===== PAGE 27 =====

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176. [UNKNOWN] paper-ii

115. Which is the major product formed in the reaction of o-chloroanisole with NaNH₂ in liquid ammonia at low temperature? (a) o-Anisidine (b) m-Anisidine (c) p-Anisidine (d) o-chloroaniline

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177. [UNKNOWN] paper-ii

116. Predict the major product formed in the following reaction: (a) (b) (c) (d) ===== PAGE 28 =====

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178. [UNKNOWN] paper-ii

117. Consider the following reaction: The major product 'P' of Wagner-Meerwein rearrangement is:

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179. [UNKNOWN] paper-ii

118. The major product formed in the following reaction is: (a) (b) (c) (d)

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180. [UNKNOWN] paper-ii

119. Which one of the following is NOT a characteristic of neighboring group participation effect? (a) An unusual stereochemical result (b) An unexpectedly fast rate of reaction (c) Formation of a rearranged product (d) The extent of neighboring group participation increases with increasing the nucleophilicity of nucleophile in solvolysis reaction ===== PAGE 29 =====

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181. [UNKNOWN] paper-ii

120. In which of the following solvents the neighboring group rate enhancement by phenyl group in the solvolysis of $\text{PhCH}_2\text{CH}_2\text{OTs}$ is maximum? (a) EtOH (b) HCOOH (c) CF_3COOH (d) CH_3COOH

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